Pharmacokinetics (PK) ADME Processes
Practice Quiz
Some items may have more than one answer.

Goals of Practice Quizzes:
1. To help students identify what they know and what they understand by applying pharmacology knowledge to clinical case vignettes.
2. To help students get used to thinking through the "Board-style" case vignettes and questions. NYITCOM exams, COMLEX and USMLE test students' medical knowledge using these types of questions.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices.
Spaced retrieval, mixing up topics, elaborating, and drawing figures, diagrams, and flow charts are learning techniques for successful learning

My recommendation: Home in on the information you need to answer the question, which, at this time, is related to drug pharmacodynamics. *Don't let yourself get distracted by the background information unless it is relevant for answering the question*.

HOW CAN I FIND THE ANSWERS? "Search" my Notes Handout. (You do not need to read the whole document if you do not wish to.)

All answers can be found by doing a "search" of my Notes Handout *or* the textbook chapters designated in the resources of each question (links provided).

- **Pharmacology bridges physiology, pathophysiology, and clinical practice.**

THE ANSWER KEY WILL BE POSTED THE FRIDAY NOVEMBER 21.

Questions are below.
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1. Why is it important for clinicians to understand fundamental pharmacology principles? Your own reflection.

2. A pharmaceutical company developed a drug for the treatment of glioblastoma (brain tumor) that can be taken orally. It is a prodrug well absorbed from the gastrointestinal (GI) tract by passive diffusion. It is nonezymatically converted to the active anticancer metabolite, MTIC, which produces high systemic blood levels (100% bioavailability). It readily penetrates the blood-brain barrier and distributes to the site of action. MTIC is not a substrate of any plasma membrane proteins. The drug is excreted mainly in the urine as inactive compounds.
Which one of the following options BEST describes the physicochemical characteristic that permits the movement of the active metabolite within the body to reach the target of action?

A. The drug has a chiral center.
B. The drug has a high lipid:water coefficient.
C. The drug forms electrostatic bonds with host structures.
D. The drug is a small molecule chemical.
E. The drug is a substrate of solute carrier (SLC) transporters.

Hint: Page 3 of Notes Handout.

3. Methamphetamine is weak base with a pKa of ~10. Approximately 33% of the dose is excreted in the urine as unchanged (unmetabolized) drug.
i) Which of the following would REDUCE the concentration of the drug in the excreted urine?

A. Urine pH 5.5
B. Urine pH 7.0

ii) Name and write out the well-known equation that explains your answer?
Hint: Pages 4-5 on the Notes Handout

4. A 55-year-old woman takes a diuretic for management of hypertension. The drug blocks sodium reabsorption through a specific ion channel in renal epithelial cells. Sodium and water are excreted (natriuresis and diuresis), lowering blood volume, which lowers blood pressure.

The physiological function of the specific ion channel is to allow movement of sodium down its electrochemical gradient into the epithelial cell, which is then reabsorbed into the bloodstream. Chloride (Cl^-) is drawn along using the energy from the sodium ion electrochemical gradient.

The question and options are on the next page.

Which one of the following best describes the mechanism of the Cl^- movement?

A. Passive diffusion
B. Aqueous diffusion
C. Paracellular transport
D. Facilitated diffusion
E. Secondary active transport
F. Primary active transport

Hint: Notes p7-9

5. Referring to the vignette above, it is stated that chloride (Cl^-) is drawn along using the energy from the sodium ion electrochemical gradient.
i) By what mechanism is the electrochemical gradient generated?
ii) What category of membrane transport generates the electrochemical gradient?

6. A 52-year-old man diagnosed with atrial fibrillation is provided an e-prescription for an anticoagulant called apixaban to reduce formation of blood clots due to the stasis of blood in the left atrium. The drug is a substrate of the proteins encoded by the *ABCB1* gene and *CYP3A4* gene.
If the patient were to require antimicrobial therapy with a drug that inhibits the proteins expressed by these genes…
i) …what would be the major potential adverse event related to the anticoagulant therapy?
ii) …what is the mechanism of the drug interaction that can cause the adverse effect?
Hint: Notes p7 and p23

7. A 74-year-old man with a history of rheumatoid arthritis and a recent diagnosis of advanced pancreatic cancer presents to the ED with altered mental status and symptoms and signs of acute salicylate toxicity. His wife found an empty bottle of regular strength aspirin near him and believes he might have tried to kill himself. She notes that he has seemed depressed for about the last month since receiving his cancer diagnosis.
The patient is given respiratory and cardiovascular support, gastric lavage and activated charcoal, and glucose supplementation. Intravenous sodium bicarbonate ($NaHCO_3$) is titrated to a urine pH of 7.5 to 8. Serum salicylate and serum potassium levels are monitored.

What is a goal of sodium bicarbonate administration in this patient?
Short phrase.
Hints: Aspirin's chemical name is acetylsalicylic acid. It has a pKa of ~3.5.
Urine pH is usually slightly acidic, 6.0. (typical range: 5.5-7.5; normal range 4.5-8.0)

8. A PGY1 medical resident on his way to work in the free clinic finds an unresponsive young female in a doorway. He detects pinpoint pupils, respiratory rate of 8 breaths per minute, and needle tracks on her arms and legs (findings are consistent with opioid overdose). He quickly administers the antidote (opioid receptor antagonist), which he always carries with him. The antidote is rapidly absorbed through mucous membranes and bypasses first-pass effect. It has an onset of action in as little as 3 minutes, a peak concentration in about 20-30 minutes, and a duration of action of up to 2 hours.
i) By what route of administration was the opioid antagonist most likely administered to the unconscious patient?

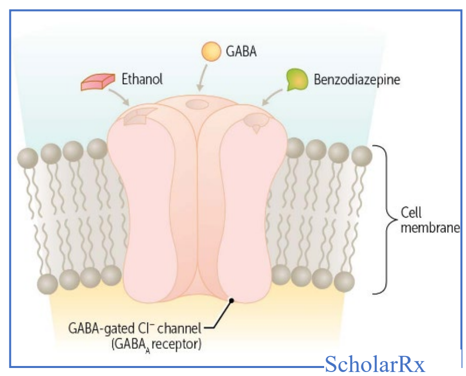
A. Intramuscular
B. Intranasal
C. Oral
D. Rectal
E. Subcutaneous
F. Transdermal

Hint: The pharmacokinetic properties will give you the clue to the correct option.
ii) What is the drug, if you happen to know?

9. A 20-year-old male being treated for methamphetamine toxicity is given a benzodiazepine for sedation. The sedative passively diffuses through cell membranes, moves across the blood-brain barrier, and binds to GABA-A receptors.
Which one of the following BEST describes the physicochemical characteristic that permits this drug's movement across physiologic barriers to reach its target of action?

A. Lipophilicity
B. Stereoisomerism
C. Structural similarity to fatty acid chains of lipid bilayer
D. Tendency to form hydrophobic bonds

Hint: Notes Part 2, page 4



10. A 4-year-old child is admitted for the treatment of bacterial meningitis with antibiotics. Which one of the following is the most appropriate route of administration in this case?

A. Intra-arterial
B. Intramuscular
C. Intraosseous
D. Intrathecal
E. Intravenous
F. Intraventricular
G. Oral

Definition: Meningitis is an inflammatory disease of the tissues surrounding the brain and spinal cord. Bacterial meningitis reflects an infection of the arachnoid mater and cerebral spinal fluid (CSF) in the subarachnoid spaces and ventricles.

Hints 1: In bacterial meningitis, the blood-brain barrier and the blood-CSF barrier become leaky due to the inflammation.
Hint 2: Bacterial meningitis is life-threatening.
Hint 3: Notes handout pp.9-13 │ PowerPoint slides 53-57.

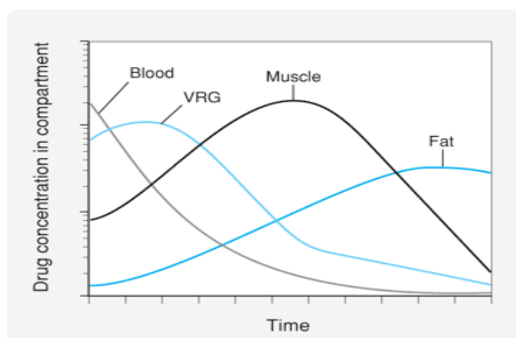
Intent of this question: To help you start thinking critically and clinically when presented with patient information.

11. A 52-year-old male being prepared for hospital discharge following treatment of deep vein thrombosis is evaluated for outpatient anticoagulant therapy. Medical history is significant for seizure disorder treated with carbamazepine (a strong CYP3A4 inducer). An oral factor Xa (pronounced ten-a) inhibitor would be effective and convenient for the patient but is contraindicated in this case.
i) What is an adverse drug event that could occur with the concomitant use of the carbamazepine and the oral factor Xa inhibitor?

A. Hemorrhagic stroke (bleeding in the brain)
B. Pulmonary embolism (thrombus breaks off and travels to the lungs)*

ii) What is the mechanism of this drug-drug interaction?
Short phrase

12. A trauma patient in the ED has a seizure. An injection of a highly lipophilic antiseizure drug with an almost immediate onset of action is administered. The duration of antiseizure effect is less about 20 minutes even though the half-life of the drug is over 24 hours.
What is the pharmacokinetic principle that best explains the short duration of action despite the long half-life of the drug?
Hint: The chart on the right



Don't forget to explain the mechanism out loud to yourself, someone else in the room, or your pet.

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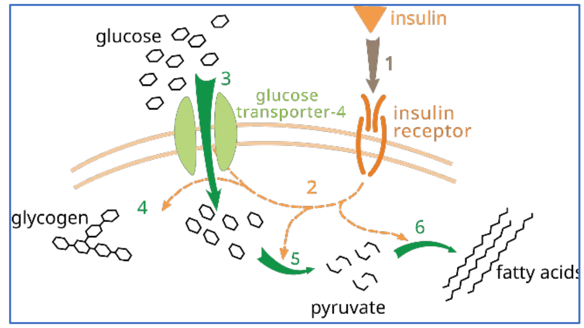
13. A 45-year-old woman injects insulin subcutaneously to manage her diabetes mellitus. Insulin is a protein that distributes in blood, diffuses into tissues, and activates its receptor tyrosine kinase on plasma membranes.
Second messenger signaling causes the GLUT4 transporters stored in intracellular vesicles to be inserted in the plasma membrane of hepatocytes, striated (skeletal and cardiac) muscle, and adipose tissue. Glucose passively enters the cell.
i) What type of transport mechanism is described?
ii) The glucose transporter is classified in what superfamily of transport proteins?
Short answers.



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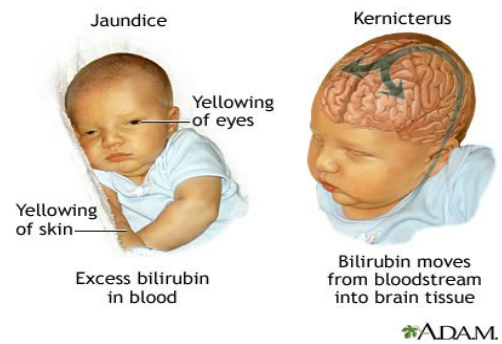
14. **Brainteaser**
A baby born 3 weeks early requires antibiotic therapy.
Sulfonamide antibiotics should be avoided in because their use can lead to accumulation of unconjugated bilirubin in the brain, which can cause encephalopathy and irreversible neurological damage.
What is the mechanism of the drug-bilirubin interaction leading to this adverse effect?



A. Blocking bilirubin hepatic excretion
B. Displacing bilirubin from albumin binding sites
C. Inhibiting the efflux pump at the blood-brain barrier
D. Inhibiting bilirubin conjugation to form the water-soluble metabolite

❖ **Information to help you identify clues and think through the question.**

Bilirubin
- Bilirubin is a highly lipophilic breakdown product of hemoglobin in aging red blood cells.
- “Unconjugated” bilirubin binds albumin in the blood for transport to the liver. Remember: Substances bound to plasma proteins do not cross physiologic barriers.
- In the liver, bilirubin is conjugated with glucuronic acid (phase 2 reaction) forming water soluble “conjugated bilirubin”, which is excreted mainly through the bile.

Newborns have immature systems:
- Newborns, especially premature babies, have immature metabolic systems, including low-functioning UDP-glucuronyltransferase (UGT), renal excretory system, and a relatively more permeable blood-brain barrier.

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Sulfonamide antibiotics PK properties:
Notice the PK property related to the information on bilirubin disposition.

A: oral / I.V.
D: Highly bound to plasma albumin
M: N-acetylation
Bioavailability: ~100%
E: urine; metabolites and unchanged drugs

15. A 53-year-old woman presents to her family physician with complaints of hot flashes that are so bothersome that they are affecting her work as an elementary school teacher and her sleep. Relevant medical history shows hysterectomy seven years ago for uterine fibroids, no current prescription medication, ibuprofen as needed for headache or musculoskeletal pain, and no known drug allergies. The patient agrees to estrogen therapy for the menopausal symptoms and for bone health. Estrogen has a high first-pass effect. The patient opts for the transdermal form of drug delivery.

i) What are some advantages of the selected route of delivery?

A. Gastric side effects are avoided.
B. Lower doses are effective.
C. Reduced dosing frequency improves patient adherence.
D. Sustained drug levels reduce peaks and troughs.
E. All of the options

ii) What property is essential for transdermal drug delivery?

16. An HIV patient's antiretroviral therapy includes drugs called oral nucleoside analogs (they are structurally similar to natural nucleosides, the building blocks of DNA and RNA). The drugs are inactive until they are activated by triphosphorylation by host kinases within the infected white blood cells where they act to prevent key viral processes.

Based on this description, the parent compound (the drug in the pills the patient swallows) is called a/an:

A. Antagonist
B. Inverse agonist
C. Prodrug
D. Xenobiotic

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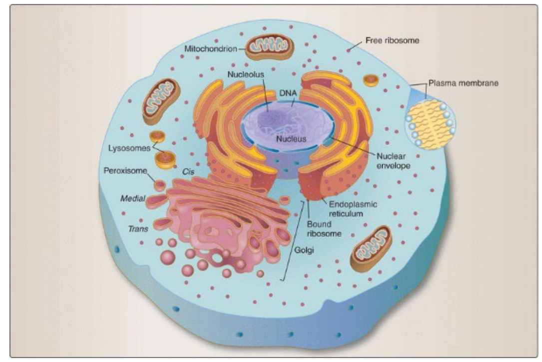
Updated August 2025

17. An adolescent female is treated for whooping cough (pertussis) with the antibiotic clarithromycin. The drug undergoes hepatic biotransformation via the cytochrome P450 monooxygenase system (CYP3A4 to be precise).
In what region of the hepatocyte does this reaction occur?



A. Endoplasmic reticulum
B. Golgi apparatus
C. Lysozymes
D. Mitochondria
E. Plasma membrane
F. Ribosomes

18. A 22-year-old male with schizophrenia resistant to first-line antipsychotic agents is admitted to the psychiatric hospital and initiated on treatment with clozapine, which has been shown to be effective for this indication. The drug is effective but considered a last-line agent due to its toxicities apparently caused by a toxic metabolite: life-threatening agranulocytosis, myocarditis, seizures. The patient smokes two packs of cigarettes per day (10 pack years) but will not be able to smoke while in the psychiatric hospital. Because the patient is a smoker, the initial dosage of the drug will be adjusted upward according to evidence-based recommendations and readjusted as needed. The patient will be closely monitored for signs and symptoms of toxicity.
Which of the following is the primary metabolic enzyme family involved in this drug interaction?

A. CYP1A family
B. CYP2B family
C. CYP2C family
D. CYP2D family
E. CYP3A family

Definitions:
Agranulocytosis: Marked decrease in granulocytes, a type of white blood cells of the innate immune system
Myocarditis: Inflammation of the heart muscle
Seizures: Sudden uncontrolled electrical disturbance in the brain.

19. A 70-year-old person (pronouns he/him/his) with recurrent metastatic colorectal cancer is evaluated for treatment with a particular chemotherapy drug that is converted to an active metabolite. This metabolite is inactivated by an enzyme that catalyzes the transfer of glucuronic acid from cofactor UDP-glucuronic acid to the active molecule. The patient's medical history reveals that he has a genetic disorder (Gilbert's syndrome), in which a gene mutation causes a deficient activity of the metabolic enzyme necessary for catalyzing the inactivation of this drug.
What class of enzymes inactivates this drug? (Short answer)

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20. A 57-year-old woman with a long history of hypertension presents to her cardiologist with complaints of fever, muscle and joint pain, and a rash on her face. Included in her medication regimen is a vasodilator metabolized by N-acetyltransferase type 2. In physical examination, a maculopapular malar rash (butterfly rash), generalized lymphadenopathy, and splenomegaly are noted. Results of the work-up reveal drug-induced lupus-like syndrome. The offending antihypertensive drug is discontinued. The symptoms and signs gradually resolve.
Which one of the following describes the biotransformation reaction that produced the toxic effect?

A. Conjugation
B. Hydrolysis
C. Oxidation
D. Reduction

Hint: Notice the mode of metabolism of the offending drug.

21. A 52-year-old woman takes conjugated estrogens for the management of menopausal vasomotor symptoms (hot flashes). Estrogen metabolism is a complex process that includes the formation of sulfate and glucuronide conjugates, which are excreted via the biliary system. When she eats a meal, the gallbladder releases its contents, including the conjugated metabolites, into the duodenum. The metabolites travel through the small intestine to the ileum, where they are hydrolyzed by bacteria in the gut flora. The reconstituted estrogen is then passively reabsorbed, which contributes to the plasma concentration of the hormone.
What pharmacokinetic process is described?
Short answer.

22. A critically ill patient receiving I.V. gentamicin (antibiotic) for a serious gram-negative bacterial infection complains of ringing in her ears and dizziness; nystagmus is noted on physical exam. The signs and symptoms are characteristic of gentamicin toxicity. Her gentamicin blood levels are found to be in the toxic range.
Gentamicin properties:

- Gentamicin is a small molecule, highly polar drug.
- It is not absorbed from the GI tract, therefore given only parenterally.
- Only a small percentage of the dose binds to plasma proteins.
- It is not metabolized.
- It is a small molecule that passes through the glomerulus.

Definition of nystagmus: Repetitive uncontrolled eye movements.

i) Impairment of what organ could lead to the toxic serum levels?
Hint: The answer can be discerned from the drug's pharmacokinetic properties.

ii) What product in serum is a useful biomarker of the function of this body system?

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Hint: Page 27 on the Notes Handout
Short answers.

23. A 65-year-old man is treated for non-Hodgkin lymphoma with a chemotherapy regimen that includes methotrexate, which has potentially serious toxicities and a narrow therapeutic window. The patient is counseled on how to recognize the adverse effects and the drug interactions to avoid. Included is avoidance of NSAIDs use, which can inhibit the secretion of methotrexate into the proximal tubule, which would increase the risk of methotrexate toxicity.
Methotrexate PK properties:
A: IV for the treatment of cancer
D: moderate plasma protein binding (~50%)
M: not metabolized by CYPs or conjugation
E: renal glomerular filtration and secretion; ~90% of the dose is excreted in urine as unchanged drug
Definition, NSAID: nonsteroidal anti-inflammatory drug, such as ibuprofen (Advil, Motrin) and naproxen (Aleve)
What is the likely pharmacokinetic mechanism of the methotrexate-NSAIDs interaction?

A. Allosteric potentiation
B. Competitive inhibition of OAT
C. Inhibition of metabolism
D. Ion trapping

Hint: Think about what can happen when the portals are blocked.

24. Excretion of unchanged nicotine is pH-dependent. At a urinary pH of 7 or greater, excretion of unchanged nicotine is approximately 2%, whereas an average of 23% is recovered unchanged at a urinary pH of less than 5.
Chemically, what type of drug is nicotine?
Short answer

25. A 38-year-old woman complains to her doctor that she has been feeling sad, hopeless, and irritable for about 6 months. She has lost interest in everything. She relates that her family and co-workers have been urging her to get help. After a full workup, she is given a diagnosis of major depressive disorder. An appointment is made with a mental health counselor and an electronic prescription (e-prescription) is submitted for an antidepressant that inhibits the reuptake of serotonin. The drug is taken orally once daily. It is 95% plasma protein bound.
If such a large percentage of the drug is protein bound, how does it achieve and maintain adequate concentrations at the site of action in the brain?
Hint: Notes page 16.